

## Technology Stack

### Project Title:

**Electricity Consumption Analysis Dashboard using Tableau and Flask**

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### Introduction

The Technology Stack describes all software, programming languages, frameworks, visualization tools, databases, and platforms used to develop the Electricity Consumption Analysis Dashboard. These technologies work together to collect, analyze, visualize, and present electricity consumption data through an interactive web application.

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### Technology Stack Used

| Technology         | Purpose                                  |
|--------------------|------------------------------------------|
| Python             | Backend programming language             |
| Flask              | Web framework used to create the website |
| Tableau Public     | Dashboard and Story visualization        |
| HTML5              | Structure of web pages                   |
| CSS3               | Styling the website                      |
| JavaScript         | Navigation and interactivity             |
| CSV Dataset        | Stores electricity consumption data      |
| Git                | Version control                          |
| GitHub             | Project repository hosting               |
| Visual Studio Code | Source code editor                       |
| Tableau Desktop    | Creating dashboards and stories          |
| Web Browser        | Running the Flask website                |

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### Software Requirements

| Software        | Version              |
|-----------------|----------------------|
| Windows 10 / 11 | Operating System     |
| Python 3.10+    | Programming Language |

| Software           | Version               |
|--------------------|-----------------------|
| Flask              | Latest Stable Version |
| Tableau Desktop    | 2026.2                |
| Visual Studio Code | Latest Version        |
| Git                | Latest Version        |
| GitHub             | Online Repository     |
| Google Chrome      | Latest Version        |

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## Programming Languages

### Python

- Flask backend
- Routing webpages
- Running the web application

### HTML

- Home Page
- Dashboard Page
- Story Page
- About Project Page

### CSS

- Styling
- Layout
- Colors
- Responsive Design

### JavaScript

- Navigation
  - User Interface
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## Visualization Tool:

### Tableau Public

Used for

- Creating Dashboard
  - Creating Story
  - Publishing Dashboard
  - Publishing Story
  - Sharing Public URLs
  - Embedding Dashboard inside Flask Website
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## **Web Framework:**

### **Flask**

Used for

- Hosting webpages
  - Connecting HTML pages
  - Routing pages
  - Running local web server
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## **Version Control:**

### **Git**

Used for

- Tracking project changes
  - Managing source code
  - Commit history
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## **Repository Hosting:**

### **GitHub**

Used for

- Uploading project
  - Sharing source code
  - Maintaining project files
  - Project documentation
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## **Dataset Format:**

CSV

Contains:

- State
  - Region
  - Sector
  - Date
  - Electricity Consumption
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## **Project Architecture**

CSV Dataset



Tableau Desktop



Dashboard & Story



Publish to Tableau Public



Embed in Flask Website



User Views Dashboard

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## **Advantages of Selected Technologies**

### **Python**

- Easy to learn
- Powerful libraries
- Fast development

### **Flask**

- Lightweight
- Simple routing
- Easy deployment

## **Tableau**

- Interactive dashboards
- Rich visualizations
- Easy publishing

## **GitHub**

- Version control
- Collaboration
- Free repository hosting

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## **Conclusion**

The selected technology stack provides an efficient and user-friendly solution for developing an Electricity Consumption Analysis Dashboard. Python and Flask manage the web application, Tableau creates interactive visualizations, and GitHub ensures proper project management and version control.